# Chapter 8, Problem 11P

## Problem

The natural response of an *RLC* circuit is described by the differential equation

$$\frac{d^2v}{dt^2} + 2\frac{dv}{dt} + v = 0$$

for which the initial conditions are $v(0) = 10$ V and $dv(0)/dt = 0$. Solve for $v(t)$.

## Step-by-step solution

**Step 1** of 8

Consider the following differential equation of *RLC* circuit:

$$\frac{d^2v}{dt^2} + 2\frac{dv}{dt} + v = 0 \quad \cdots\cdots (1)$$

Write the generalised expression for *RLC* circuit.

$$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{1}{LC}v = 0 \quad \cdots\cdots (2)$$

Here, $v$ is the voltage across the capacitor.

**Step 2** of 8

Compare the equations (1) and (2).

$$\frac{R}{L} = 2$$

$$\frac{1}{LC} = 1$$

Calculate the damping factor $\alpha$.

$$\alpha = \frac{R}{2L} = \frac{\left(\frac{R}{L}\right)}{2}$$

Substitute 2 for $\left(\frac{R}{L}\right)$.

$$\alpha = \frac{2}{2} = 1$$

**Step 3** of 8

Calculate the undamped natural frequency, $\omega_0$.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \sqrt{\frac{1}{LC}}$$

Substitute 1 for $\frac{1}{LC}$.

$$\omega_0 = \sqrt{1} = 1$$

**Step 4** of 8

The roots of the characteristic equation are,

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -1 \pm \sqrt{1^2 - 1^2} = -1$$

The roots of the critically damped system are,

$$s_1 = s_2 = -\alpha = -1$$

**Step 5** of 8

Since $\alpha = \omega_0$, the circuit is critically damped and the roots are real and equal. The response of the system is,

$$v(t) = (A_1 + A_2 t)e^{-\alpha t}$$

Substitute 1 for $\alpha$.

$$v(t) = (A_1 + A_2 t)e^{-t} \quad \ldots\ldots (3)$$

Substitute 0 for $t$.

$$v(0) = (A_1 + A_2 \times 0)e^{-0}$$
$$v(0) = A_1$$

Since $v(0) = 10$, substitute 10 for $v(0)$ in above expression.

$$A_1 = 10$$

**Step 6** of 8

Recall equation (3).

$$v(t) = (A_1 + A_2 t)e^{-t}$$

Differentiate with respect to *t*.

$$\begin{aligned}\frac{dv(t)}{dt} &= \frac{d}{dt}\left(A_1 e^{-t} + A_2 t e^{-t}\right) \\ &= A_1\left(-e^{-t}\right) + A_2\left(-te^{-t} + e^{-t}\right)\end{aligned}$$

Substitute 0 for $t$.

$$\begin{aligned}\frac{dv(0)}{dt} &= A_1\left(-e^{-0}\right) + A_2\left(-0 \times e^{-0} + e^{-0}\right) \\ &= -A_1 + A_2\end{aligned}$$

**Step 7** of 8

Since $\frac{dv(0)}{dt} = 0$, substitute 0 for $\frac{dv(0)}{dt}$ in above expression.

$$\begin{aligned}0 &= -A_1 + A_2 \\ A_2 &= A_1\end{aligned}$$

Substitute 10 for $A_1$.

$$A_2 = 10$$

**Step 8** of 8

Recall equation (3).

$$v(t) = (A_1 + A_2 t)e^{-t}$$

Substitute 10 for $A_1$ and $A_2$.

$$v(t) = (10 + 10t)e^{-t}\ \text{V}$$

Therefore, the expression for the voltage, $v(t)$ is $\boxed{(10+10t)e^{-t}\ \text{V}}$.